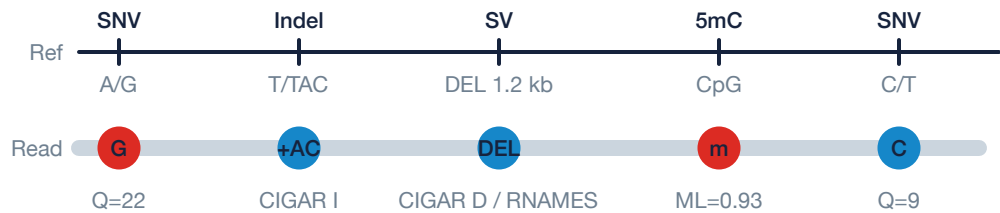


a Allele observations extracted from one alignment

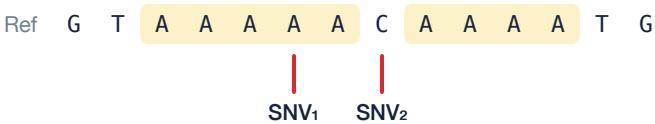


Observation quality used for edge weighting

- SNV** Phred base quality; edge weight 1 if both bases $\geq$ Q12, else 0.1
- Indel** CIGAR I/D starting at the site; fixed high quality; tandem-repeat indels flagged
- SV** caller read list $\rightarrow$ ALT (high quality); unlisted reads $\rightarrow$ REF (lower quality)
- 5mC** modcall read lists, strand-matched; fixed high quality

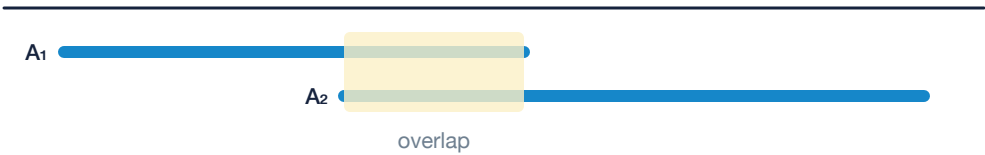
See Supplementary Fig. 4 for the copy-number-aware filter applied after these steps.

b Homopolymer SNV filter (nanopore)



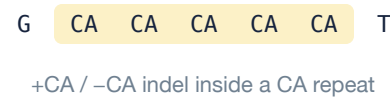
Both SNVs in homopolymers of length ≥ 3 and ≤ 2 bp apart
 $\rightarrow$ downstream SNV removed from all reads

c Overlapping-alignment filter



Two alignments of one read; overlap / combined span ≥ 0.2
 $\rightarrow$ the shorter alignment (A₁) is discarded

d Tandem-repeat indel flag



Flagged indels stay in the graph but vote with weight 0.1
and contribute 0.1 to read haplotype assignment